Homework 3
Network Dynamics and Learning

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1 Influenza H1N1 2009 Pandemic in Sweden

1.1 Epidemic on a Known Graph

In this part, we simulate the spread of an epidemic on a known network using the SIR (Susceptible–Infected–Recovered) model. The population is represented by a k-regular undirected graph, where each node corresponds to an individual and each edge represents a social contact between two individuals.

We first generate a symmetric k-regular graph with $|V| = 500$ nodes and degree $k = 4$. The epidemic dynamics are then simulated on this network using the SIR model. At each time step, every node can be in one of three states: Susceptible (S), Infected (I), or Recovered (R).

A susceptible node becomes infected depending on the number of its infected neighbors. The infection probability is given by

$$P\left(X_i(t+1) = I \mid X_i(t) = S\right) = 1 - (1-\beta)^m, \tag{1}$$

where m is the number of infected neighbors of node i and β is the transmission rate.

An infected node recovers with probability ρ at each time step, defined as

$$P\left(X_i(t+1) = R \mid X_i(t) = I\right) = \rho. \tag{2}$$

In our simulations, we set $\beta = 0.3$ and $\rho = 0.7$. Initially, 10 nodes are randomly selected to be infected, while all other nodes start in the susceptible state. The epidemic is simulated for 15 time steps (weeks).

To reduce randomness, the simulation is repeated $N = 100$ times. For each week, we record:

- the average number of newly infected individuals,
- the average total number of susceptible individuals,
- the average total number of infected individuals,

- the average total number of recovered individuals.

Finally, the results are averaged over all simulations and plotted to show the evolution of the epidemic over time.

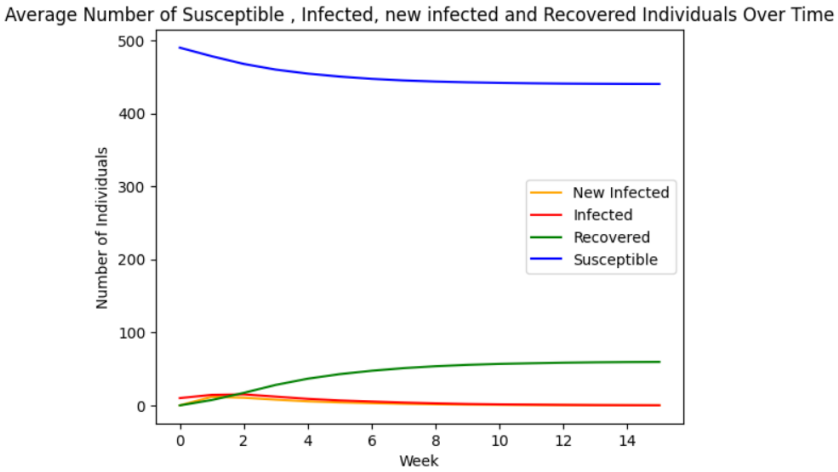

Figure 1: Average number of susceptible, infected, and recovered individuals over time (known k-regular graph).

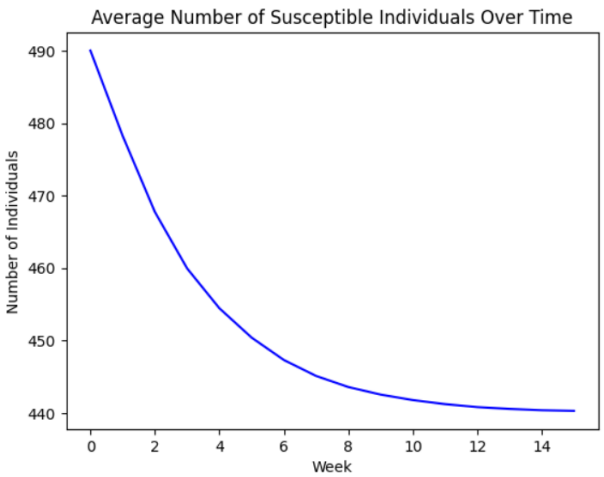

Figure 2: Alternative visualization of S, I, R evolution (known k-regular graph).

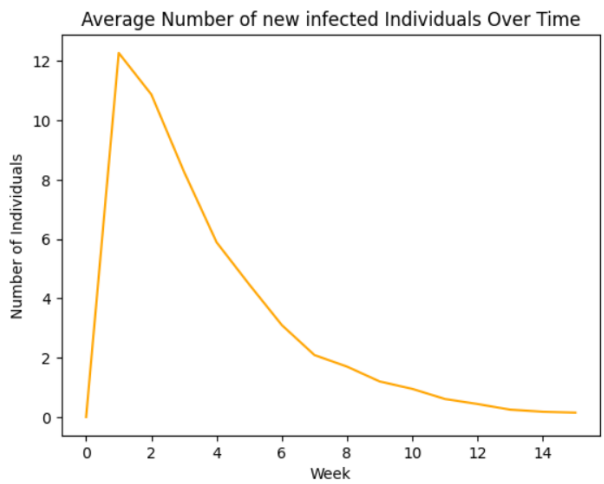

Figure 3: Average number of new infections per week over $N = 100$ simulations (known k-regular graph).

Figures 1–2 show the average number of susceptible, infected, and recovered individuals over time. Initially, almost the entire population is susceptible. As time progresses, the number of susceptible individuals decreases gradually, the number of infected individuals remains low and declines quickly, and the number of recovered individuals increases until it stabilizes.

Figure 3 shows the average number of new infections per week over $N = 100$ simulation runs. The number of new infections is high at the beginning of the simulation and then decreases over time. This indicates that the disease does not spread for a long period due to a high recovery rate.

1.2 Generate a Random Graph

In this part, we generate a random graph using the Preferential Attachment model. This model captures the idea that new nodes prefer to connect to existing nodes with higher degrees, which is commonly observed in real-world networks.

The graph generation starts from a complete graph with $(k + 1)$ nodes. New nodes are then added one by one until the desired number of nodes n is reached. Each new node connects to approximately $k/2$ existing nodes. The probability that a new node connects to an existing node i depends on the degree of node i relative to the total degree of the network.

Formally, the probability of connecting to node i at time t is given by

$$P(i) = \frac{w_i(t-1)}{\sum_{j \in V_{t-1}} w_j(t-1)}, \tag{3}$$

where $w_i(t-1)$ denotes the degree of node i at time $t-1$.

To handle cases where k is odd, the number of selected neighbors alternates between $\lfloor k/2 \rfloor$ and $\lceil k/2 \rceil$. This ensures that the average degree of the network remains close to the desired value k while preserving randomness. As a result, the structure of the generated graph depends on the chosen values of k and the total number of nodes n.

1.3 Simulate a Pandemic Without Vaccination

Using the random graph generated by the Preferential Attachment model, we simulate the spread of an epidemic without vaccination using the SIR model. Each node represents an individual, and edges represent social contacts through which the disease can spread.

The simulation is performed under the following assumptions:

$$|V| = 500, \quad k = 6, \quad \beta = 0.3, \quad \rho = 0.7,$$

with 10 initially infected nodes. The epidemic is simulated for 15 weeks.

At each time step, susceptible nodes may become infected depending on the number of infected neighbors, while infected nodes recover with probability ρ. To reduce stochastic effects, the simulation is repeated 100 times, and the results are averaged.

During the simulation, we record:

- the average number of newly infected individuals per week,
- the average total number of susceptible individuals per week,
- the average total number of infected individuals per week,
- the average total number of recovered individuals per week.

The results show that even without vaccination, the epidemic eventually dies out. In the next section, we study how vaccination affects the spread and control of the pandemic.

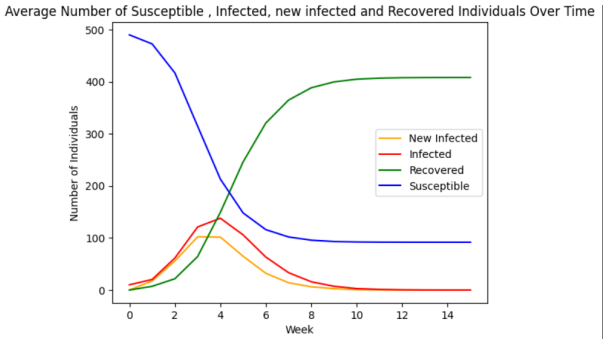

Figure 4: SIR evolution on preferential attachment graph without vaccination.

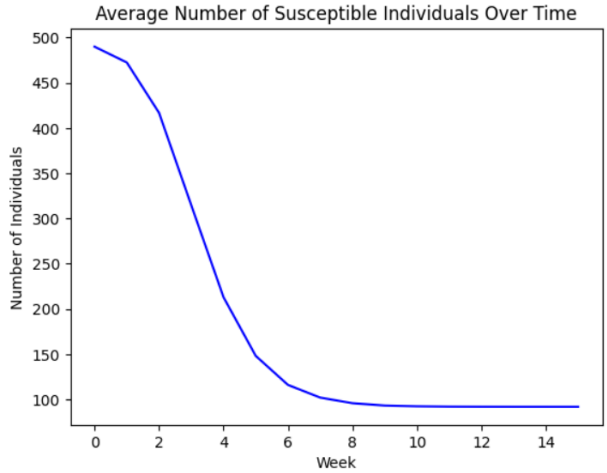

Figure 5: New infections per week on preferential attachment graph without vaccination.

In a preferential attachment graph, some nodes have many connections. If one of these nodes becomes infected, it can infect many others quickly. This causes a high number of new infections at the beginning, followed by a rapid decrease as these nodes recover.

Even though the average degree is $k = 6$, some nodes have far more connections. Compared to $k = 4$, the disease spreads more easily in the preferential attachment graph. As a result, more individuals are infected and recovered.

1.4 Simulate a Pandemic with Vaccination

In this part, we extend the previous epidemic simulation by incorporating vaccination into the SIR model. In addition to the susceptible (S), infected (I), and recovered (R) states, a vaccinated (V) state is introduced to represent individuals who are immunized and cannot become infected.

The simulation is performed on a graph generated using the Preferential Attachment model. The initial assumptions are the same as in the previous section:

$$|V| = 500, \quad k = 6, \quad \beta = 0.3, \quad \rho = 0.7,$$

with 10 initially infected nodes. Each simulation runs for 15 weeks and is repeated 100 times to reduce stochastic effects.

Vaccination is applied gradually according to a predefined vaccination schedule,

$$\text{Vacc}(t) = [0, 5, 15, 25, 35, 45, 55, 60, 60, 60, 60, 60, 60, 60, 60],$$

which represents the cumulative percentage of the population vaccinated over time. At each week, the number of new vaccinations is computed based on the increase in this schedule and scaled to the total population size. Individuals to be vaccinated are randomly selected from the population that has not yet been vaccinated, regardless of their current epidemiological state.

During the simulation, we record:

- the average number of newly infected individuals per week,

- the average total number of susceptible individuals per week,
- the average total number of infected individuals per week,
- the average total number of recovered individuals per week.

The results show that vaccination significantly reduces the number of infections compared to the scenario without vaccination. Although the effect is moderate for this population size, the impact of vaccination becomes more pronounced in larger networks.

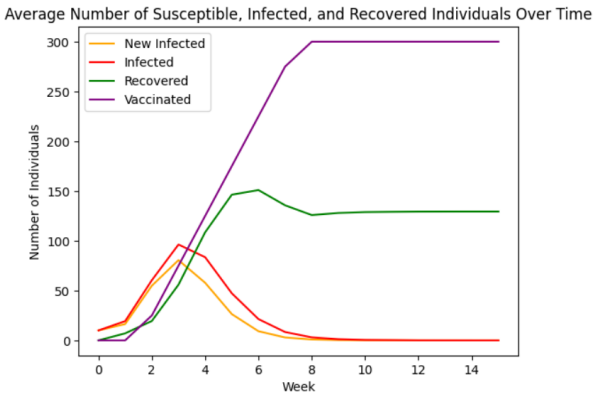

Figure 6: SIRV evolution on preferential attachment graph with vaccination.

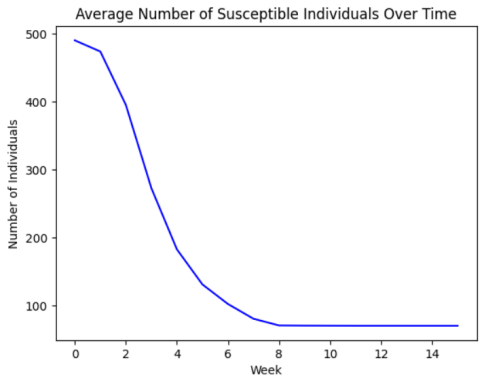

Figure 7: Weekly new infections and vaccination trend (preferential attachment graph with vaccination).

New infections rise at the beginning, peak around week 3, and then decrease steadily, becoming almost zero after weeks 8–9. The number of newly vaccinated individuals increases during the first weeks and reaches its peak between weeks 1 and 6; after week 7–8 it remains constant. The total number of vaccinated individuals grows quickly and stabilizes at about 300 around weeks 7–8. At the same time, the number of susceptible individuals drops sharply and then remains nearly constant. Infected individuals increase early, peak around week 3, and decline to almost zero, while the number of recovered individuals increases and stabilizes over time.

1.5 The H1N1 Pandemic in Sweden 2009

The goal of this part is to estimate suitable values for the epidemic model parameters k, β, and ρ by comparing simulation results with real epidemic data. The parameters are chosen such that the simulated number of infected individuals is as close as possible to the real weekly infection data.

To measure the accuracy of the model, the Root Mean Square Error (RMSE) is used. The parameter set that produces the smallest RMSE is considered the best fit. After identifying the optimal parameters, the epidemic is simulated again to validate the model and to study the behavior of the system over time. In addition, the total number of susceptible (S), infected (I), recovered (R), and vaccinated (V) individuals is analyzed throughout the simulation.

The study is based on a stochastic SIRV model on a preferential attachment network with $|V| = 934$ nodes. The simulation runs for 15 weeks and starts with one randomly infected individual.

To estimate the parameters, a grid search is performed around the initial values $k = 10$, $\beta = 0.3$, and $\rho = 0.6$. For each combination of parameters, the epidemic is simulated $N = 10$ times. The average number of newly infected individuals per week is then compared to the real data using the RMSE criterion. The parameter set with the lowest RMSE is selected as the best estimate.

The optimal parameters obtained are $k = 8$, $\beta = 0.2$, and $\rho = 0.8$, with an RMSE value of 4.38. These parameters are then used in 100 additional simulations to further validate the results and analyze the epidemic dynamics. These results show that the calibrated model is able to capture the overall trend of the H1N1 pandemic in Sweden, and that parameter optimization using RMSE provides a reasonable approximation of real epidemic dynamics.

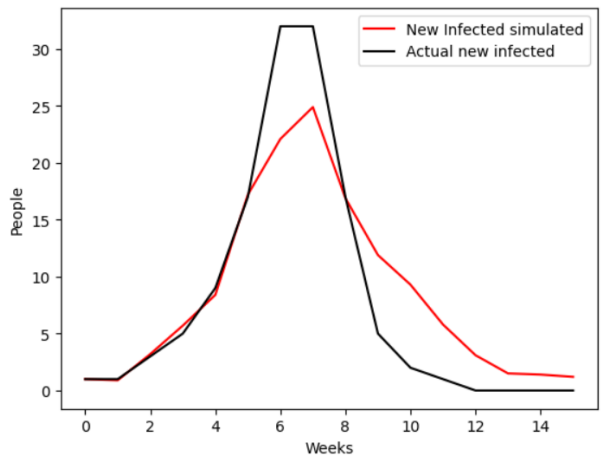

Figure 8: Comparison between real data and simulated new infections (best-fit parameters).

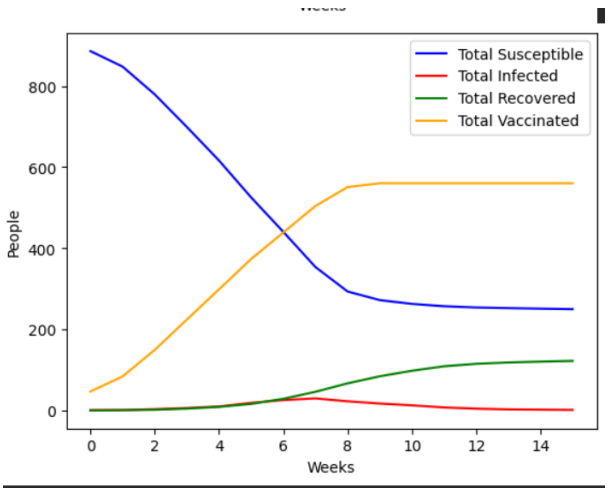

Figure 9: S, I, R, V evolution under the best-fit parameters.

2 Network games and dynamics

We consider a network game with $n = 3$ players, action set $\mathcal{A} = \{-1, +1\}$, and a partition of players into two groups:

$\mathcal{V}_1 = \{1, \ldots, n_1\}$ (coordination players), $\mathcal{V}_2 = \{n_1+1, \ldots, 3\}$ (anti-coordination players).

For an action profile $x = (x_1, x_2, x_3) \in \{-1, +1\}^3$, utilities are

$$u_i(x) = \begin{cases} \frac{1}{2} \sum_{j \neq i} |x_i + x_j|, & i \in \mathcal{V}_1, \\ \frac{1}{2} \sum_{j \neq i} |x_i - x_j|, & i \in \mathcal{V}_2. \end{cases} \tag{4}$$

The tasks are:

(a) For each $n_1 \in \{0, 1, 2, 3\}$, compute all *pure Nash equilibria*.

(b) Consider *asynchronous best response* dynamics as a Markov chain on $\{-1, +1\}^3$ (one randomly chosen player updates at each time step, tie-breaking uniformly). For the initial condition

$$X(0) = (+1, -1, +1),$$

represent the transition graph and compute long-run/limit behavior for:

(b1) $n_1 = 3$ (all coordination),

(b2) $n_1 = 0$ (all anti-coordination).

(c) Consider *noisy best response* (logit) dynamics and characterize the behavior as noise vanishes (inverse-noise parameter $\beta \to \infty$).

2.1 Model simplification and interpretation

Because $x_i, x_j \in \{-1, +1\}$, we have the identities:

$$\frac{1}{2}|x_i + x_j| = \begin{cases} 1, & x_i = x_j, \\ 0, & x_i \neq x_j, \end{cases} \qquad \frac{1}{2}|x_i - x_j| = \begin{cases} 0, & x_i = x_j, \\ 1, & x_i \neq x_j. \end{cases}$$

Hence (4) simplifies to:

$$u_i(x) = \begin{cases} \sum_{j \neq i} \mathbf{1}\{x_i = x_j\}, & i \in \mathcal{V}_1, \\ \sum_{j \neq i} \mathbf{1}\{x_i \neq x_j\}, & i \in \mathcal{V}_2, \end{cases} \tag{5}$$

where $\mathbf{1}\{\cdot\}$ is the indicator function.

Interpretation. With $n = 3$, each player compares its action to the other two players:

- A coordination player ($i \in \mathcal{V}_1$) receives payoff $0, 1$, or 2 depending on whether it matches $0, 1$, or 2 of the other players.
- An anti-coordination player ($i \in \mathcal{V}_2$) receives payoff $0, 1$, or 2 depending on whether it differs from $0, 1$, or 2 of the other players.

2.2 Pure Nash equilibria for all n_1

A profile $x^\star$ is a pure Nash equilibrium (PNE) if no player can improve utility by unilateral deviation:

$$u_i(x^\star) \geq u_i(a_i, x^\star_{-i}) \quad \forall i,\ \forall a_i \in \{-1, +1\}.$$

Using (5), for $n = 3$ we obtain the PNE sets below (these were computed and verified by exhaustive enumeration in Python).

2.2.1

section$n_1 = 3$ (all coordination) All players want to match the others. The only PNE are the two consensus profiles:

$$(+,+,+), \qquad (-,-,-).$$

2.2.2 $n_1 = 2$ (players 1–2 coordinate, player 3 anti-coordinates)

Players 1–2 prefer to match; player 3 prefers to differ. The PNE are:

$$(+,+,-), \qquad (-,-,+).$$

2.2.3 $n_1 = 1$ (player 1 coordinates, players 2–3 anti-coordinate)

The computed PNE set contains four profiles:

$$(+,+,-), \quad (+,-,+), \quad (-,+,-), \quad (-,-,+).$$

2.2.4 $n_1 = 0$ (all anti-coordination)

Every player wants to differ from the others. With $n = 3$ and binary actions, full pairwise disagreement cannot happen simultaneously, so equilibria are exactly the *six split states*:

$$(+,+,-),\ (+,-,+),\ (-,+,+),\ (-,-,+),\ (-,+,-),\ (+,-,-).$$

2.2.5 Consistency check with the computed output

The Python output for PNE matches the sets above:

- $n_1 = 3$: $(1,1,1)$ and $(-1,-1,-1)$.
- $n_1 = 2$: $(1,1,-1)$ and $(-1,-1,1)$.
- $n_1 = 1$: $(1,1,-1)$, $(1,-1,1)$, $(-1,1,-1)$, $(-1,-1,1)$.
- $n_1 = 0$: the six split states.

2.3 Asynchronous best response dynamics as a Markov chain

We consider a discrete-time asynchronous best response (BR) dynamic:

1. At each time t, choose one player $i \in \{1, 2, 3\}$ uniformly at random (probability 1/3).
2. Player i updates its action to a best response to $x_{-i}(t)$.
3. If there is a tie (both actions are best responses), break ties uniformly.

This defines a Markov chain $\{X(t)\}_{t\geq 0}$ on the 8 states $\{-1, +1\}^3$ with transition matrix P.

2.3.1 Transition graph for $n_1 = 3$ (all coordination)

The computed transition graph (from the output) is:

- $(1, 1, 1)$ is absorbing.
- $(-1, -1, -1)$ is absorbing.
- Mixed states: if a chosen player is the *minority*, its unique best response is to flip to match the majority; if a chosen player is part of the majority, it can be indifferent, creating self-loops and multiple outgoing edges.

Example interpretation (key). From $(1, -1, 1)$:

$$(1, -1, 1) \to (1, 1, 1) \text{ with prob } 1/3,$$

because if player 2 is selected (prob 1/3), its best response is to switch from -1 to $+1$ and match both others.

2.3.2 Long-run behavior for $n_1 = 3$ starting from $X(0) = (1, -1, 1)$

Because $(1, 1, 1)$ and $(-1, -1, -1)$ are absorbing and all other states are transient, the chain is absorbed almost surely in one of the two consensus states. The computed absorption probabilities are:

$$\mathbb{P}(\text{absorb at } (1, 1, 1) \mid X(0) = (1, -1, 1)) = \frac{2}{3}, \quad \mathbb{P}(\text{absorb at } (-1, -1, -1) \mid X(0) = (1, -1, 1)) = \frac{1}{3}.$$

2.3.3 Transition graph for $n_1 = 0$ (all anti-coordination)

In the all anti-coordination case, the two consensus states are transient and the six split states form a single recurrent communicating class.

2.3.4 Long-run behavior for $n_1 = 0$ starting from $X(0) = (1, -1, 1)$

The chain converges to the unique stationary distribution supported on split states:

$$\pi(x) = \frac{1}{6} \text{ for each split state } x, \qquad \pi(1,1,1) = \pi(-1,-1,-1) = 0.$$

2.4 Noisy best response (logit) dynamics and vanishing noise limit

We implement a logit (noisy) best response dynamic: player i chooses action $a \in \{-1, +1\}$ with probability

$$\mathbb{P}(x_i = a \mid x_{-i}) = \frac{\exp(\beta\, u_i(a, x_{-i}))}{\exp(\beta\, u_i(+1, x_{-i})) + \exp(\beta\, u_i(-1, x_{-i}))}, \tag{6}$$

where $\beta > 0$ is the inverse noise parameter. For any finite β, the chain is ergodic and has a unique stationary distribution π_β.

2.4.1 $n_1 = 3$ (coordination): limit as $\beta \to \infty$

As $\beta \to \infty$, π_β concentrates equally on the two consensus states:

$$\pi_\beta \xrightarrow[\beta\to\infty]{} \frac{1}{2}\delta_{(1,1,1)} + \frac{1}{2}\delta_{(-1,-1,-1)}.$$

2.4.2 $n_1 = 0$ (anti-coordination): limit as $\beta \to \infty$

As $\beta \to \infty$, the stationary distribution converges to uniform over split states:

$$\pi_\beta \xrightarrow[\beta\to\infty]{} \text{Uniform distribution on the six split equilibria.}$$

2.5 Conclusions

The computed outputs are coherent with theory. Pure Nash equilibria match the exhaustive enumeration, and the Markov-chain behavior under best response and logit dynamics is consistent with coordination versus anti-coordination incentives.